

Topic 1: How the Internet and Web Work

Learning Objectives

- Understand the difference between the Internet and the World Wide Web
- Know how DNS resolves domain names to IP addresses
- Understand the HTTP request-response cycle
- Know how browsers receive and render web pages

1.1 The Internet vs the World Wide Web

The **Internet** is a global network of computers connected to each other using cables, fiber optics, and wireless signals. It is the physical infrastructure — the "roads" of the digital world.

The **World Wide Web (WWW)** is a service that runs on top of the Internet. It is a collection of web pages and resources (HTML, CSS, images, videos) that you access through a browser. Think of the Internet as the highway system, and the Web as the cars and destinations on those highways.

Other services that also use the Internet (but are not the Web):

- Email (SMTP / IMAP)
- File Transfer (FTP)
- Video calls (WebRTC)
- Online gaming

1.2 Clients and Servers

Every website interaction involves two sides:

- **Client** — Your browser (Chrome, Firefox, Safari). It requests and displays content.
- **Server** — A computer somewhere that stores web files and sends them when requested.

When you visit a website, your browser (client) sends a request to the server. The server responds with HTML, CSS, and JavaScript files. Your browser then renders them into the page you see.

1.3 IP Addresses and DNS

Every device on the Internet has a unique **IP address** (e.g., `192.168.1.1`). Servers also have IP addresses, but we use domain names like `google.com` instead of memorizing numbers.

DNS (Domain Name System) works like a phone book for the Internet:

1. You type `www.google.com` in your browser
2. Your computer asks a DNS server: "What is the IP address of google.com?"
3. The DNS server replies: "It is `142.250.80.46` "
4. Your browser connects to that IP address

1.4 HTTP and HTTPS

HTTP (HyperText Transfer Protocol) is the set of rules defining how browsers and servers communicate.

HTTPS is the secure version — all data is encrypted using TLS/SSL so that nobody can intercept it. Always look for the padlock icon in your browser. Modern websites must use HTTPS.

Common HTTP methods:

- `GET` — Request a web page or resource
- `POST` — Send data to the server (e.g., login form)
- `PUT` — Update existing data
- `DELETE` — Remove data

1.5 The Request-Response Cycle

Here is what happens when you type a URL and press Enter:

1. Browser checks its cache for a saved copy
2. DNS lookup — domain name converted to IP address
3. Browser opens a TCP connection to the server
4. Browser sends an HTTP GET request
5. Server responds with a status code and content
6. Browser receives HTML, then requests CSS and JS files
7. Browser parses and renders the full page

HTTP Status Codes:

Code	Meaning
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200	OK — Request succeeded
301	Moved Permanently — Redirect
404	Not Found — Page does not exist
500	Internal Server Error — Server-side problem

1.6 How Browsers Render a Web Page

Once the HTML file arrives, the browser follows these steps:

1. **Parse HTML** — builds the DOM (Document Object Model)
2. **Parse CSS** — builds the CSSOM (CSS Object Model)
3. **Combine DOM + CSSOM** — creates the Render Tree
4. **Layout** — calculates sizes and positions
5. **Paint** — draws pixels onto the screen

Key Takeaways

- The Internet is physical infrastructure; the Web is a service on top of it
- Every website interaction is a client request and a server response
- DNS translates domain names to IP addresses
- HTTPS encrypts communication for security
- Browsers render pages by parsing HTML, CSS, then painting the screen

Reference Materials

- YouTube: How The Web Works (Big Picture) — <https://www.youtube.com/watch?v=hJHvdBISxug>
- YouTube: Internet vs World Wide Web — <https://www.youtube.com/watch?v=-pKW2Ju12NA>
- MDN: How the Web Works — https://developer.mozilla.org/en-US/docs/Learn/Getting_started_with_the_web/How_the_Web_works

Quiz

Question 1: What does DNS stand for?

- A) Dynamic Network Service
- B) Domain Name System
- C) Digital Network Standard
- D) Data Naming Structure

Correct Answer: B — DNS (Domain Name System) translates human-readable domain names like google.com into the IP addresses computers use to connect.

Question 2: Which HTTP status code means the requested page was not found?

- A) 200
- B) 301
- C) 404
- D) 500

Correct Answer: C — HTTP 404 means "Not Found". The server received the request but could not locate the requested resource.
